Automatic Stenosis Detection in Coronary CT Angiography

A Two-Stage Approach Combining Deep Learning Segmentation with Computational Geometry

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Abstract

Coronary artery disease (CAD) is a leading cause of mortality worldwide. Coronary CT angiography (CCTA) is a widely adopted non-invasive imaging modality for assessing the presence and severity of coronary artery stenosis, but manual analysis is time consuming, subject to inter-reader variability, and requires specialized expertise. We present an automated system for detecting and quantifying coronary artery stenosis from CCTA scans using a two-stage approach. The first stage employs a supervised 3D U-Net trained on the ImageCAS dataset to segment the coronary vessel tree, followed by a post-processing pipeline that removes spurious fragments while preserving anatomically significant vessel components. The second stage applies unsupervised computational geometry methods—centerline extraction, artery labeling, local radius estimation, and segment-based stenosis quantification—to produce clinically interpretable stenosis measurements. This deliberate separation of learned segmentation from geometric analysis avoids black-box severity prediction and yields results that are directly comparable to clinical reporting standards. The complete pipeline is deployed as an interactive desktop application. We evaluate the full pipeline against radiologist ground truth from Ziv Medical Center using threshold based classification metrics and ROC analysis.

1 Introduction

Coronary artery disease (CAD) is a condition characterized by the narrowing of coronary arteries due to atherosclerotic plaque buildup, restricting blood flow to the heart muscle. Stenosis, the abnormal narrowing of a blood vessel, is the hallmark finding that clinicians assess to determine CAD severity and guide treatment decisions. Coronary CT angiography (CCTA) has become a widely adopted non invasive imaging modality for assessing the presence and degree of coronary artery stenosis. However, manual analysis of CCTA scans is time-consuming, subject to inter reader variability, and requires specialized expertise.

In recent years, deep learning models have shown promising results in automating tasks in medical imaging. The goal of this project is to develop an AI-based system that can automatically detect and quantify coronary artery stenosis from CCTA images.

1.1 Our Approach

We propose a two stage system that deliberately separates vessel segmentation from stenosis quantification:

1. **Supervised segmentation**: A 3D U-Net model segments the coronary vessel tree from preprocessed CCTA volumes, producing a binary vessel mask.
2. **Unsupervised geometric analysis**: Computational geometry methods extract the vessel centerline, classify segments by coronary artery system, estimate local lumen radii, and quantify stenosis as relative narrowing within anatomically defined vessel segments.

This separation offers several advantages. The geometric stenosis measurement is fully interpretable every detected narrowing can be traced to specific radius measurements along the vessel centerline. The stenosis formula, severity thresholds, and reference diameter computation are all transparent and adjustable. Furthermore, the segmentation model can be trained on datasets with vessel masks alone (without stenosis annotations), while stenosis quantification requires no training at all.

1.2 System Requirements

The system takes as input 3D CCTA volumes in DICOM or NIfTI format and produces:

- A binary segmentation mask of the coronary vessel tree.
- A vessel centerline with local radius estimates and artery labels at each point.
- Per segment stenosis measurements with clinical severity classifications.
- An interactive 3D visualization, multi-planar 2D slice views with mask overlay, and a structured JSON clinical report.

2 Data

Two complementary datasets are used in this project, each serving a distinct role in the pipeline.

2.1 ImageCAS Dataset

The ImageCAS dataset [Zeng et al., 2023] is a publicly available collection of approximately 1,000 CCTA volumes with corresponding voxel level coronary vessel annotations. Each volume is provided as a NIfTI file with an accompanying binary mask delineating the coronary artery tree.

ImageCAS is used exclusively for training, validating, and testing the segmentation model. The dataset is split into training (750 volumes, 75%), validation (150 volumes, 15%), and test (100 volumes, 10%) sets using a fixed random seed for reproducibility.

2.2 Ziv Medical Center Dataset

The Ziv Medical Center dataset consists of 25 CCTA volumes acquired on a Philips scanner as part of routine clinical care. Unlike ImageCAS, the Ziv dataset does not include voxel level vessel annotations. Instead, each case is accompanied by a structured radiology report that specifies stenosis severity and anatomical location for each affected coronary artery segment.

These reports serve as the ground truth for evaluating the complete stenosis detection pipeline. A dedicated PDF parser extracts structured findings (artery name, stenosis severity,

and associated details) from the clinical reports, enabling automated comparison with system predictions.

The Ziv dataset is used for two purposes: (1) evaluating domain transfer of the segmentation model trained on ImageCAS, and (2) quantitative evaluation of the full pipeline against clinical ground truth.

3 Preprocessing

Each CCTA volume undergoes a standardized preprocessing pipeline to ensure consistency across patients and datasets. The pipeline is applied identically to both ImageCAS and Ziv data, which proved critical for successful domain transfer.

3.1 Orientation Standardization

Raw CCTA volumes may be stored in different spatial orientations depending on the scanner and acquisition protocol. We reorient all volumes to a common RAS+ (Right-Anterior-Superior) coordinate system using the orientation information encoded in the NIfTI affine matrix. This step proved critical: an initial orientation mismatch between the ImageCAS training data and Ziv clinical data (where Ziv data was originally processed to LAS orientation) caused complete model failure on clinical data, despite strong training performance. Standardizing both datasets to RAS resolved this issue entirely.

3.2 HU Windowing

Hounsfield Unit (HU) values outside the range of interest are clipped to reduce the influence of irrelevant structures. We apply a window of $[\mathrm{HU}_{\min}, \mathrm{HU}_{\max}] = [-50, 700]$:

$$x_{\text{clipped}} = \begin{cases} -50 & \text{if } x < -50 \\ x & \text{if } -50 \leq x \leq 700 \\ 700 & \text{if } x > 700 \end{cases} \tag{1}$$

This window captures the contrast-enhanced coronary vasculature while excluding bone (> 1000 HU) and air (< -500 HU). The specific window $[-50, 700]$ was selected based on empirical analysis of the preprocessed data characteristics a key insight was that training decisions should be guided by the statistics of the processed data rather than raw scan properties. The effect of alternative HU windows on segmentation performance is examined empirically in Section **??**.

3.3 Isotropic Resampling

Raw CCTA scans typically have anisotropic voxel spacing, with the in plane resolution (e.g., 0.35×0.35 mm) being much finer than the slice thickness (e.g., 0.5–1.0 mm). We resample all volumes to isotropic spacing of $0.5 \times 0.5 \times 0.5$ mm using linear interpolation (order 1), which corresponds to trilinear interpolation in 3D.

For a volume with original spacing (s_x, s_y, s_z) and target spacing (t_x, t_y, t_z), the new grid dimensions are:

$$N'_d = \left\lfloor N_d \cdot \frac{s_d}{t_d} + 0.5 \right\rfloor, \quad d \in \{x, y, z\} \tag{2}$$

where N_d and N'_d are the original and resampled dimensions along axis d.

Image volumes are resampled using linear interpolation (order 1). For binary segmentation masks, we apply linear interpolation followed by thresholding:

$$m_{\text{resampled}}(\mathbf{v}) = \begin{cases} 1 & \text{if } \tilde{m}(\mathbf{v}) > 0.3 \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

where $\tilde{m}$ is the linearly interpolated mask. This approach was a critical discovery: nearest-neighbor interpolation (the standard choice for binary masks) caused approximately 82% vessel voxel loss during resampling due to the thin, elongated geometry of coronary vessels. Linear interpolation with a threshold of 0.3 preserves vessel connectivity while maintaining mask accuracy.

3.4 Z-Score Normalization

After HU windowing and resampling, voxel intensities are normalized using z-score normalization with clipping:

$$\tilde{x} = \begin{cases} -3 & \text{if } z < -3 \\ z & \text{if } -3 \leq z \leq 3 \\ 3 & \text{if } z > 3 \end{cases}, \quad z = \frac{x - \mu}{\sigma} \tag{4}$$

where μ and σ are the per-volume mean and standard deviation computed over the windowed image, and the result is clipped to the range $[-3, 3]$ to suppress outliers. This normalization ensures that all volumes share a comparable intensity distribution regardless of scanner calibration or contrast agent concentration.

3.5 Storage Format

Preprocessed volumes are stored in Zarr format with Blosc compression, which provides efficient random access to 3D subvolumes during training. Each Zarr store contains the image array, the corresponding mask (for ImageCAS), and metadata attributes including original spacing, applied transformations, normalization parameters, and preprocessing version.

3.6 Summary of Preprocessing Parameters

Table 1 summarizes the complete preprocessing configuration.

Table 1: Preprocessing pipeline parameters.

Step	**Parameter**	**Value**
Orientation	Target	RAS+
HU windowing	Range	$[-50, 700]$
Resampling	Target spacing	$0.5 \times 0.5 \times 0.5$ mm
Resampling	Image interpolation	Linear (order 1)
Resampling	Mask interpolation	Linear + threshold at 0.3
Normalization	Method	Z-score
Normalization	Clipping range	$[-3, 3]$
Storage	Format	Zarr v3 with Blosc compression

4 Segmentation Model

The first stage of the pipeline is a deep learning model that produces a binary vessel mask from the preprocessed CCTA volume. Few model variants were trained on the ImageCAS dataset and applied to both ImageCAS validation data and Ziv clinical data.

4.1 Architecture

We use a 3D U-Net [Çiçek et al., 2016] implemented via the MONAI framework [MONAI Consortium, 2024]. The encoder–decoder architecture consists of five resolution levels with channel

dimensions $(32, 64, 128, 256, 512)$, downsampling strides of $(2, 2, 2, 2)$, two residual units per level, batch normalization, and dropout of 0.1 in deeper levels. The model takes single channel input and produces a single channel output, followed by a sigmoid activation to obtain voxel level vessel probabilities.

Formally, for an input volume $X \in \mathbb{R}^{D \times H \times W}$, the model produces:

$$\hat{M}(X) = \sigma\big(f_\theta(X)\big) \in [0, 1]^{D \times H \times W} \tag{5}$$

where f_θ denotes the U-Net parameterized by θ and σ is the sigmoid function. A threshold τ is applied at inference time to obtain the binary mask; the optimal value of τ depends on the loss function used during training (see Section 5).

4.1.1 Attention U-Net Variant

In addition to the standard U-Net, we implement an Attention U-Net variant that incorporates two attention mechanisms inspired by recent work on coronary artery segmentation [Wu et al., 2024, Dong et al., 2024]:

- **Parallel channel and spatial attention** in the encoder, which computes channel wise and spatial importance maps simultaneously (rather than sequentially as in CBAM) and fuses them via element wise summation.
- **Attention gates** in the decoder skip connections, which learn to suppress irrelevant encoder features before concatenation, allowing the decoder to focus on vessel relevant information.

Both model variants share the same five level encoder–decoder structure with identical channel dimensions. In practice, the Attention U-Net produces marginally higher Dice scores and more structurally coherent vessel graphs, particularly in complex branching regions and edge cases where the standard U-Net tends to fragment. However, it requires substantially longer training time, higher GPU memory, and more powerful hardware for local inference, making it less practical for deployment on consumer grade machines.

4.2 Loss Function

Coronary vessel segmentation presents an extreme class imbalance problem: vessels typically occupy only 0.15–0.20% of the total volume, yielding a background to foreground ratio of approximately 600:1. We address this with a combined loss function:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{Dice}} \cdot \mathcal{L}_{\text{Dice}} + \lambda_{\text{BCE}} \cdot \mathcal{L}_{\text{BCE}} \tag{6}$$

Several training configurations are evaluated, differing in the weighting of the loss components:

1. **DiceBCE model** (primary): $\lambda_{\text{Dice}} = 0.7$, $\lambda_{\text{BCE}} = 0.3$. Balances volumetric overlap and boundary precision.
2. **Dice-only model**: $\lambda_{\text{Dice}} = 1$, $\lambda_{\text{BCE}} = 0$. Optimizes exclusively for volumetric overlap. In practice, this configuration yields slightly higher precision than DiceBCE with comparable overall Dice, as the absence of per voxel BCE gradients produces a more conservative segmentation boundary. The difference between Dice only and DiceBCE is primarily a shift in the precision-recall tradeoff rather than a substantial change in overall segmentation quality.
3. **BCE-only model**: $\lambda_{\text{Dice}} = 0$, $\lambda_{\text{BCE}} = 1$. Optimizes exclusively for per voxel cross-entropy, producing higher recall at lower precision, amenable to aggressive post-processing cleanup at a higher decision threshold, will discuss this more in later sections.

4.2.1 Dice Loss

The Dice loss directly optimizes the overlap between predicted and ground truth masks:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum_{i=1}^{N} \hat{m}_i \cdot m_i + \epsilon}{\sum_{i=1}^{N} \hat{m}_i + \sum_{i=1}^{N} m_i + \epsilon} \tag{7}$$

where $\hat{m}_i \in [0, 1]$ is the predicted probability for voxel i, $m_i \in \{0, 1\}$ is the ground truth label, N is the total number of voxels in the patch, and $\epsilon = 10^{-5}$ is a smoothing constant. Dice loss is inherently robust to class imbalance because it operates on a ratio rather than a sum of per voxel errors.

4.2.2 Binary Cross-Entropy with Positive Weighting

The BCE component provides per voxel gradients that improve boundary precision:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^{N} \Big[w_+ \cdot m_i \log(\hat{m}_i) + (1 - m_i) \log(1 - \hat{m}_i) \Big] \tag{8}$$

where $w_+ = 5.0$ is a positive class weight that mildly boosts foreground gradients without destabilizing training. This weight is intentionally much smaller than the true class ratio ($\sim$600), as excessively large weights cause over segmentation. The BCE loss is computed on logits (before sigmoid) for numerical stability.

The rationale for this combination is as follows: Dice loss handles class imbalance naturally through its ratio formulation and drives the model to optimize for volumetric overlap. BCE provides per voxel supervision that refines vessel boundaries. Earlier experiments with Focal loss [Lin et al., 2017] led to training collapse, as the down weighting of confident predictions suppressed gradients at vessel boundaries where the model is inherently uncertain.

4.3 Training Configuration

Training is performed on Google Colab Pro+ with NVIDIA A100 GPU (40/80 GB). Table 2 summarizes the training hyperparameters.

Table 2: Training configuration.

Parameter	**Value**
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
Learning rate	10^{-4}
Weight decay	10^{-5}
LR scheduler	Cosine annealing
Batch size	4
Patch size	$160 \times 160 \times 160$ voxels
Foreground sampling ratio	0.70
Data augmentation	Intensity scaling, shifting, Gaussian noise, contrast
Mixed precision	FP16 with gradient scaling
Maximum epochs	150
Early stopping patience	30 epochs
Validation interval	Every 5 epochs

4.3.1 Patch-Based Training

Due to the large size of CCTA volumes, the model is trained on randomly sampled 3D patches. Three patch sizes were evaluated: 128^3, 160^3, and 192^3. The 160^3 patch size achieved the

best segmentation performance, offering a favorable balance between spatial context and training efficiency, 128^3 patches provided insufficient context for larger vessel structures, while 192^3 patches did not improve results and increased memory requirements. To ensure sufficient vessel representation in each batch, 70% of patches are sampled with centers placed on foreground (vessel) voxels, while the remaining 30% are sampled uniformly. Foreground voxel locations are precomputed and cached for each volume to avoid repeated mask scanning during training.

4.3.2 Inference

At inference time, the full volume is processed using sliding window inference with 50% overlap and Gaussian weighting for smooth blending at patch boundaries. This ensures that every voxel in the volume receives a prediction, with boundary artifacts minimized by the overlapping, weighted aggregation. The sigmoid output probabilities are retained as a float16 probability map alongside the thresholded binary mask, enabling threshold sweeps without re-running inference.

4.3.3 Data Augmentation

Several data augmentation strategies were evaluated during training, including intensity scaling, intensity shifting, Gaussian noise injection, and random contrast adjustment. After systematic testing, we found that augmentation consistently degraded segmentation performance on this task. Models trained without augmentation achieved higher Dice scores than their augmented counterparts across all configurations tested. This finding is consistent with observations reported by other groups working on coronary artery segmentation, where the precise intensity characteristics of contrast enhanced vessels appear to be important training signals that augmentation disrupts. **Consequently, the final models are trained without data augmentation.**

5 Post-Processing Pipeline

The raw segmentation mask often contains spurious isolated fragments that arise from false positive activations near nonvessel structures (e.g., calcified valves, image artifacts). A post processing pipeline runs after segmentation and before centerline extraction to remove these fragments while preserving both coronary artery systems.

5.1 Pipeline Steps

The post processing module implements a configurable sequence of cleanup operations:

1. **Small component removal**: Connected components smaller than a minimum size threshold are discarded.
2. **Gap bridging** (optional): Small components within a maximum distance of the main vessel trees are connected via cylindrical bridges.
3. **Distance-based fragment removal**: Components not among the top-K protected trees are removed if their nearest point to any protected tree exceeds $d_{\max}$.
4. **Final small component cleanup**: A second pass of minimum-size filtering removes fragments introduced by intermediate steps.

The top-K protected trees mechanism is central to this pipeline. The K largest connected components (by voxel count) that each exceed a minimum fraction $f_{\min}$ of total vessel volume are designated as "protected" they are never removed by any filtering step. This ensures that both the left coronary system and the right coronary system are always preserved, even when post processing aggressively removes noise. A safety floor (minimum total voxels) prevents pathological cases from being reduced to empty masks.

5.2 Probability-Cache Infrastructure

To evaluate different threshold and post-processing configurations without repeating expensive GPU inference, we built a probability cache: the raw float16 sigmoid output for each case is stored on disk alongside the binary mask and ground truth. This infrastructure enabled a systematic sweep over decision thresholds ($\tau \in \{0.30, 0.35, 0.40, 0.45, 0.50\}$ for the DiceBCE model; $\tau \in \{0.50, 0.60, 0.70, 0.80\}$ for the BCE-only model) and post-processing configurations.

5.3 Tuning Results

All threshold and post-processing experiments were conducted on the validtion and testing sets of ImageCAS cases (seed = 42) with the ground truth masks available for Dice and recall computation.

The optimal post-processing configuration for the DiceBCE model is: $K = 3$ protected trees, $d_{\max} = 5$ mm distance filter, minimum component size = 200 voxels, with hole filling, smoothing, bridging, and shape filtering all disabled.

5.3.1 BCE-Only Model

The BCE-only model produces over-segmented predictions at default threshold ($\tau = 0.50$): vessel voxel counts are 1.4–1.8× the ground truth, confirming the expected high-recall, low-precision behavior. Raising the decision threshold compensates for this.

The optimal BCE-only post-processing configuration is tighter: $K = 3$ protected trees, $d_{\max} = 3$ mm, minimum component size = 1,000 voxels. Distance filtering is the primary contributor (+0.54–0.63% Dice).

6 Centerline Extraction

The binary segmentation mask is transformed into a structured vessel representation suitable for stenosis analysis through a series of morphological and graph based operations.

6.1 Mask Cleanup

Before centerline extraction, the binary mask undergoes a lightweight cleanup pass within the centerline module. Since the post-processing pipeline (Section 5) is an optional step that may be skipped, this cleanup ensures a minimum level of mask quality regardless of whether full post-processing was applied:

1. **Small component removal**: Connected components smaller than 200 voxels are removed to eliminate isolated noise clusters that would otherwise produce spurious skeleton branches.

6.2 Skeletonization

The cleaned binary mask is reduced to a one-voxel-wide centerline using morphological skeletonization [Lee et al., 1994]. This operation iteratively removes boundary voxels while preserving the topological structure of the vessel tree, producing a medial axis representation:

$$S = \text{skeletonize}(M_{\text{clean}}) \tag{9}$$

where S is the binary skeleton and M_{clean} is the cleaned segmentation mask.

6.3 Distance Transform and Radius Estimation

The Euclidean distance transform is applied to the original (pre-skeleton) binary mask:

$$\mathrm{DT}(\mathbf{v}) = \min_{\mathbf{u} \notin M} \|\mathbf{v} - \mathbf{u}\|_2, \quad \forall \mathbf{v} \in M \tag{10}$$

For each voxel $\mathbf{v}$ inside the mask, $\mathrm{DT}(\mathbf{v})$ computes the Euclidean distance to the nearest background voxel $\mathbf{u}$, which corresponds to the radius of the largest inscribed sphere centered at $\mathbf{v}$ and thus approximates the local vessel radius. Evaluating the distance transform at skeleton points yields the radius profile along the centerline:

$$r(\mathbf{p}) = \mathrm{DT}(\mathbf{p}) \cdot s, \quad \mathbf{p} \in S \tag{11}$$

where $s = 0.5$ mm is the isotropic voxel spacing, converting the distance transform output from voxel units to physical millimeters. Since the skeleton passes through the approximate medial axis of the vessel, the resulting $r(\mathbf{p})$ traces the vessel's radius profile along its length.

To suppress noise from segmentation boundary irregularities, the raw radius estimates are smoothed using a Gaussian filter convolved along the arc length of each vessel segment:

$$\tilde{r} = G_\sigma * r \tag{12}$$

with $\sigma = 1.0$ mm, where $*$ denotes one-dimensional convolution along the segment arc length. This smoothing reduces sensitivity to local segmentation noise while preserving meaningful radius variations such as stenotic narrowing.

This radius estimation approach has several known limitations. The distance transform measures the radius of the largest inscribed sphere at each point, which assumes a circular vessel cross-section; for elliptical or irregular lumens, particularly near bifurcations, this yields the minimum rather than the mean radius. Segmentation errors propagate directly into radius estimates—local over- or under-segmentation at a stenosis site can corrupt the stenosis score even when the rest of the segment is accurately segmented. The skeletonization algorithm does not guarantee exact geometric centering, introducing a small systematic underestimation bias. Finally, Gaussian smoothing with $\sigma = 1.0$ mm may attenuate very short stenoses (< 2 mm in extent), motivating the inclusion of σ as a tunable hyperparameter (Section **??**).

6.4 Graph Construction

Skeleton voxels are classified by their local connectivity:

- **Endpoints**: degree = 1 (terminal points of vessel branches).
- **Regular points**: degree = 2 (interior points along a vessel segment).
- **Bifurcation points**: degree ≥ 3 (branching points where vessels split).

Nearby bifurcation points (within 3 voxels of each other) are merged into single bifurcation nodes to avoid artificial oversplitting of vessel segments at complex branching regions.

The classified skeleton points define a graph $G = (V, E)$ where nodes V are endpoints and bifurcations, and edges E are vessel segments connecting them through chains of regular points.

6.5 Segment Extraction

The vessel tree is decomposed into individual segments, where each segment is a path between two graph nodes (endpoint to bifurcation, bifurcation to bifurcation, or endpoint-to-endpoint). Each segment carries:

- An ordered sequence of 3D centerline coordinates.

- Smoothed radius values at each point.
- The arc length from segment start to each point.
- The segment's total length in millimeters.

Segments shorter than 8 mm are excluded from stenosis analysis unless they connect two bifurcation points (in which case they may represent short but clinically relevant vessel regions). Dead-end branches shorter than 3 mm are pruned as spurious artifacts of skeletonization.

Optionally, centerline points are resampled to uniform spacing (0.3 mm) using spline interpolation to ensure consistent sampling density for downstream analysis.

7 Artery Labeling

Each vessel segment extracted from the centerline graph is assigned an anatomical label (Left Coronary, Right Coronary, or LM/LAD/LCx if the left system can be further subdivided) and a regional descriptor (proximal, mid, or distal). This labeling enables stenosis findings to be reported by artery system and location, matching the format of clinical radiology reports.

7.1 Previous Approaches and Their Failures

Two earlier approaches were developed and abandoned after testing revealed fundamental limitations.

The first approach attempted to label arteries using aorta detection and ostia localization: detecting the aorta as the brightest large blob in the CT volume, locating ostia where coronary vessels contact the aortic wall, rooting vessel trees via breadth-first search (BFS) from ostia, and classifying branches using direction vectors relative to the aorta. This approach succeeded on only 7 of 42 segments (16.7%) on a test case. The primary failure modes were: (1) aorta detection fails on z-score normalized volumes where the aorta's intensity is not distinctive, and (2) BFS tree traversal only reaches segments connected within the same component, so vessel fragmentation caused by the segmentation model leaves large portions of the vessel tree unreachable.

The second approach for LAD/LCx subdivision built a rooted tree from the full point-by-point centerline graph using BFS and searched for bifurcations where child branches had radius exceeding 35% of the parent radius. This failed because the point-by-point centerline graph is too dense—every centerline voxel is a graph node, so true anatomical bifurcations (where two *segments* diverge) are buried among thousands of degree-2 nodes. The BFS tree could not reliably identify the LAD/LCx split point because the topological signal was lost in the noise of the dense graph. Additionally, the tree-building process traversed through both left and right coronary components when they shared any nearby nodes, producing a linear chain through the entire vessel tree with no visible bifurcations.

7.2 Component-Based Approach

The current approach abandons aorta detection entirely and instead uses connected components of the vessel mask as the fundamental labeling unit. The module requires only the vessel mask and centerline data as input—the CT volume is not needed, reducing memory usage and simplifying the pipeline.

7.2.1 Step 1: Map Segments to Components

`scipy.ndimage.label()` with 26-connectivity is applied to the vessel mask, assigning each foreground voxel to a connected component. Each centerline segment is then mapped to a component by looking up the component label at its midpoint voxel. If the midpoint misses the mask (due

to discretization), three additional sample points (at 25%, 50%, and 75% of arc length) are used with majority voting.

7.2.2 Step 2: Merge Nearby Components

Vessel fragmentation from the segmentation model can split a single coronary system into multiple disconnected components. To address this, a union-find merge step is applied before classification. For each pair of significant components, a KDTree nearest-neighbor query computes the minimum surface-to-surface distance. Components whose surfaces are closer than a distance threshold $d_{\text{merge}} = 8.0$ mm are merged into a single component. This threshold exploits the anatomical separation between coronary systems: fragments from the same system are typically 3–15 mm apart, while the left-to-right inter-system distance is 30–50 mm. After merging, the segment-to-component mapping is recomputed on the updated labeled mask.

7.2.3 Step 3: Classify Components as Left or Right

Components are classified by size and spatial position. Components above a minimum size threshold (5% of total vessel volume) are designated as "significant." The significant components are sorted by their mean x-coordinate in the RAS array. A critical correction was required here: in the array index space used by the segmentation mask, lower x-indices correspond to the patient's *right* side, not the left. The component with the lowest mean x-index is therefore classified as the Right Coronary, and the component with the highest mean x-index as the Left Coronary. Intermediate significant components are assigned to the nearest system by x-proximity. Small components (below the 5% threshold) are labeled "Minor Vessel." If only a single significant component exists, it receives the generic label "Coronary" since left–right distinction is not possible.

7.2.4 Step 4: Assign Segment Labels

Every centerline segment inherits the label of its mapped component. Confidence scores are assigned based on the component's fraction of total vessel volume:

- $\geq 15\%$ of total volume: confidence 0.9 (major coronary system).
- $\geq 5\%$ of total volume: confidence 0.7 (significant but not dominant).
- $< 5\%$ of total volume: confidence 0.6 (minor vessel).
- Unmapped segment (centerline does not intersect mask): confidence 0.4.

7.2.5 Step 5: LAD/LCx Subdivision via Junction Graph

For left-system segments, a best-effort subdivision into Left Main (LM), Left Anterior Descending (LAD), and Left Circumflex (LCx) is performed. Unlike the failed earlier approach that operated on the dense point-by-point centerline graph, this method constructs a *junction graph* where nodes are segment endpoints (endpoints and bifurcations) and edges are entire vessel segments. This abstraction makes anatomical bifurcations directly visible as graph nodes with degree ≥ 3.

The algorithm proceeds as follows:

1. **Build junction graph.** For each left-system segment, the start and end node IDs define edges in a segment-level adjacency structure. Each junction node maps to the list of segments and neighboring junctions it connects. The graph is restricted to left-system segments to prevent traversal into the right coronary during the subsequent walk.

2. **Find LM root.** The left main artery exits the aorta at the most superior point of the left coronary tree. The endpoint (degree-1 junction) with the highest z-index in the array is selected as the presumed LM root. This heuristic is robust because the LM ostium is consistently the most superior left-sided endpoint across patients, and it does not depend on radius estimates that can be unreliable near the aortic attachment.

3. **Walk from root with on-demand reconnection.** Starting from the LM root, the algorithm walks distally through the junction graph. At each step, before examining neighbors, it attempts to reconnect the current junction to nearby unvisited junctions that are within $d_{\text{reconnect}} = 8.0$ mm and reachable through the vessel mask. A candidate connection is accepted only if a straight line between the two junction positions passes entirely through foreground voxels, verified by sampling points along the line at sub-voxel resolution. This vessel-validated reconnection repairs gaps introduced by segmentation fragmentation without creating spurious cross-system connections, since the line cannot jump through empty space to an unrelated vessel. Connections are added to all valid nearby junctions at once rather than only the nearest, which is important for correctly bridging trifurcation-like configurations where the LM meets both the LAD and LCx at nearby but distinct points.

4. **Identify the LAD/LCx bifurcation.** At each junction encountered during the walk, the algorithm checks whether two or more outgoing branches qualify as "major," defined as having a subtree total segment length $\geq l_{\min} = 10.0$ mm. A depth-first traversal from each branch measures the cumulative segment length of its downstream subtree. Short dead-end branches (first segment shorter than $l_{\min}$ with a target junction that has no further real connections) are immediately classified as minor without full traversal. If fewer than two major branches exist, minor branches are absorbed into the LM path and the walk continues along the single major branch. When two or more major branches are found, that junction is identified as the LAD/LCx bifurcation point.

5. **Classify branches by direction.** The two largest-radius branches at the bifurcation are selected. For each branch, a direction vector is computed from the first 20 centerline points of the departing segment. In RAS array coordinates, the LAD courses anteriorly (toward lower y-index) while the LCx courses posteriorly (toward higher y-index). The branch with the lower y-component is labeled LAD; the other is labeled LCx. Empirical validation confirmed that y-axis separation is the reliable discriminator between LAD and LCx territories, rather than x-axis separation.

6. **Label remaining segments by position.** All left-system segments on the walk path from root to bifurcation are labeled LM. Remaining left-system segments are assigned to LAD or LCx based on their midpoint y-coordinate relative to the bifurcation point: segments with midpoint y below the bifurcation y are labeled LAD, and those above are labeled LCx. This position-based assignment is more robust than subtree-based BFS assignment in fragmented graphs, where the junction topology often causes BFS to leak across the bifurcation and misattribute segments.

7. **Neighbor connectivity correction.** A final pass checks each LAD/LCx segment against its graph neighbors. If all of a segment's neighbors (excluding LM segments) carry the opposite label, the segment is relabeled to match its neighbors. This corrects borderline cases where the y-position heuristic misclassifies a segment whose midpoint lies just on the wrong side of the bifurcation plane but which is topologically part of the other artery's subtree.

If the walk fails to find a bifurcation with two or more major branches, left-system segments retain the generic "Left Coronary" label.

7.2.6 Step 6: Region Assignment

For segments that received sub-artery labels (LM, LAD, LCx), a regional descriptor is assigned based on arc-length position within the artery. Segments belonging to the same artery are sorted by their distance from the root, and cumulative arc length is computed. The midpoint fraction of each segment determines its region:

- $< 33\%$ of total artery length: proximal.
- 33–67%: mid.
- $> 67\%$: distal.

This produces labels such as "proximal LAD" or "distal LCx," consistent with the anatomical terminology used in clinical reports.

7.3 Fallback Strategy

The labeling module is designed to degrade gracefully when information is incomplete:

Table 3: Artery labeling fallback behavior.

Scenario	Label Assigned	Confidence
No vessel mask provided	Coronary	0.5
Single significant component	Coronary	0.5
Two or more significant components	Left / Right Coronary	0.7–0.9
LAD/LCx split succeeds	LM / LAD / LCx	0.7
LAD/LCx split fails	Left Coronary (unchanged)	0.9
Unmapped centerline segment	Unknown	0.4

8 Stenosis Detection

Stenosis is quantified on a per-segment basis by comparing the minimum local radius to a reference radius that represents the expected healthy vessel caliber within that segment. A sequence of artifact filters is applied afterward to suppress false positives caused by imperfect segmentation, which would otherwise dominate the output on clinical data.

8.1 Reference Diameter Estimation

For each vessel segment, the reference radius r_{ref} is defined as the q-th percentile of the smoothed radius values along that segment:

$$r_{\text{ref}} = P_q\big(\{\tilde{r}(\mathbf{p}) : \mathbf{p} \in \text{segment}\}\big) \tag{13}$$

where P_q denotes the q-th percentile and $q = 80$ in the production configuration. The percentile was increased from an earlier value of 75 after empirical tuning: an 80th-percentile reference produces stenosis scores slightly closer to the radiologist-reported values on validation cases, because the top quartile of radii better represents the healthy caliber in the presence of minor measurement noise near the stenotic region.

Radius values near segment endpoints are excluded from the reference computation (within 2 mm of each end) to avoid the influence of bifurcation geometry, where the vessel cross-section may be distorted.

Computing reference diameters per segment rather than globally is essential. Different vessel segments have naturally different calibers (e.g., the left main artery is larger than distal

branches), and a global reference would systematically overestimate stenosis in smaller vessels and underestimate it in larger ones. This segment-based approach mirrors clinical practice, where stenosis is assessed relative to the local healthy vessel diameter.

8.2 Radius Smoothing

Before quantification, the raw radius profile is smoothed with a one-dimensional Gaussian filter to suppress voxel-level noise. A smoothing sigma of $\sigma = 0.5$ is used in the production configuration, reduced from an earlier default of 1.0. The lower sigma preserves short narrowing events that stronger smoothing would erase, which matters for clinically relevant focal stenoses. Over-smoothing was found to systematically flatten measured narrowings on validation cases, pushing stenosis scores below the clinical severity thresholds.

8.3 Stenosis Quantification

The stenosis score for each segment is computed as the relative narrowing at the point of minimum radius:

$$S = 1 - \frac{r_{\min}}{r_{\text{ref}}} \tag{14}$$

where $r_{\min} = \min_{\mathbf{p} \in \text{segment}} \tilde{r}(\mathbf{p})$ is the minimum smoothed radius within the segment.

This score ranges from 0 (no narrowing) to values approaching 1 (near-total occlusion). Negative scores, which can occur when $r_{\min} > r_{\text{ref}}$ due to statistical fluctuations, are clamped to zero.

8.4 Bifurcation Suppression

Vessel bifurcation regions are geometrically complex: the vessel cross-section is irregular and the distance transform does not accurately reflect the lumen radius at branching points. This causes spurious narrowing detections near bifurcations. To address this, centerline points within a suppression zone around each bifurcation are excluded from stenosis analysis:

$$\text{suppression zone} = \{\ \mathbf{p} : \|\mathbf{p} - \mathbf{b}\| < d_{\text{bif}} \cdot r_{\text{bif}}\ \} \tag{15}$$

where $\mathbf{b}$ is a bifurcation point, r_{bif} is the radius at that bifurcation, and $d_{\text{bif}} = 1.5$ is a scaling factor. This adaptive suppression zone is proportional to vessel size, ensuring appropriate suppression for both large proximal and small distal bifurcations.

8.5 Artifact Filters

Initial evaluation on 28 Ziv cases against radiologist ground truth revealed systematic false-positive findings caused by segmentation gaps. When the segmentation model fails to capture part of a vessel, the centerline extraction terminates at the mask boundary. The distance transform gives near-zero radius at this boundary, and the stenosis detector interprets the radius drop as severe stenosis—even though the underlying vessel is anatomically normal. Six of sixteen normal patients produced false severe findings from this mechanism before filtering. Four filters were added after the initial stenosis regions are detected to suppress these artifacts without discarding true positives.

Endpoint tapering filter. Findings whose peak location lies within $d_{\text{taper}} = 6.0$ mm of either segment endpoint are discarded as likely natural tapering or mask-boundary artifacts. This threshold was increased from an earlier value of 4.0 mm after empirical analysis showed that segmentation-edge artifacts extend 4–6 mm from the true edge, because the distance transform decays smoothly rather than dropping sharply at the mask boundary.

Radius gradient filter. The main distinction between real stenosis and segmentation gaps is the rate of radius change. Real coronary stenosis develops gradually: a moderate narrowing in a 2 mm vessel reduces the radius from approximately 1.0 mm to 0.5 mm over 3–10 mm of arc length, yielding a gradient of 0.05–0.17 mm/mm. A segmentation gap causes the radius to drop from a normal value to near-zero within 1–2 mm, producing gradients of 0.75–2.0 mm/mm. The filter computes the maximum radius gradient in a five-point window centered on the stenosis peak:

$$g_{\max} = \max_{i \in \text{window}} \left| \frac{\tilde{r}(\mathbf{p}_{i+1}) - \tilde{r}(\mathbf{p}_i)}{\ell_{i+1} - \ell_i} \right| \tag{16}$$

where ℓ_i is the arc length at point i. If $g_{\max} > g_{\max\,\text{drop}} = 0.5$ mm/mm and the minimum radius is below $r_{\text{artifact}} = 0.55$ mm, the finding is discarded. The radius guard prevents this filter from rejecting real stenoses in small vessels where low radii are anatomically plausible.

Minimum radius floor. Findings with $r_{\min} < r_{\text{floor}} = 0.25$ mm in segments longer than 20 mm are discarded. A 20 mm segment represents a major coronary artery, and a 0.25 mm radius corresponds to a sub-millimeter lumen diameter that would present as complete occlusion rather than stenosis on CTA. This filter catches extreme gap artifacts that pass the gradient check.

Extreme narrowing filter. The gradient filter misses cases where a segmentation gap produces a smoothly decaying radius profile rather than an abrupt drop. To handle these, findings with $r_{\min}/r_{\text{ref}} < 0.25$ in segments longer than 40 mm are discarded. A ratio below 0.25 corresponds to $> 75\%$ stenosis, and such severe narrowings in major vessels (> 40 mm) are rare enough that the false-positive reduction from filtering outweighs the risk of missing a true finding. This filter was added after observing false severe findings on cases where the segmentation boundary faded gradually over several millimeters, bypassing the gradient check.

Confidence gate. After all per-segment filters, a global confidence gate removes findings with confidence below $c_{\min} = 0.60$. The confidence score is a weighted combination of the stenosis percentage, segment length, distance from the nearest endpoint, and the number of stenotic points. Low-confidence findings are typically short mild narrowings near endpoints in small vessels—the cases most prone to artifacts. This threshold removes the weakest findings without affecting high-confidence clinically significant detections.

8.6 Severity Classification

Stenosis scores are mapped to clinical severity levels following standard cardiology guidelines:

Table 4: Stenosis severity classification.

Severity	Stenosis Score	Clinical Interpretation
Normal	$S < 0.25$	No hemodynamically significant narrowing
Mild	$0.25 \leq S < 0.50$	Minor narrowing, typically managed conservatively
Moderate	$0.50 \leq S < 0.70$	Significant narrowing, may require intervention
Severe	$S \geq 0.70$	Critical narrowing, intervention typically indicated

The system produces a per-segment severity classification and an artery-level label (from Section 7) that together form a finding comparable to entries in clinical radiology reports.

8.7 Stenosis Detection Parameters

Beyond the reference percentile and severity thresholds, additional parameters control detection quality. Detected stenoses shorter than 1.0 mm in extent along the centerline are discarded as likely single-voxel artifacts, and adjacent detections within 2.0 mm of each other are merged into a single finding. Table 5 summarizes the full set of stenosis detection parameters, including the artifact filter thresholds described in Section 8.5.

Table 5: Stenosis detection pipeline parameters.

Category	**Parameter**	**Value**
Reference diameter	Percentile q	80
	Exclude ends	2.0 mm
Radius processing	Smoothing sigma σ	0.5
	Min segment points	5
Bifurcation	Distance suppression	2.0 mm from bifurcation
	Radius factor	1.5× local median
Region detection	Min stenosis length	1.0 mm
	Merge distance	2.0 mm
Endpoint taper	Taper distance d_{taper}	6.0 mm
	Taper fraction	5% of segment length
Gradient filter	Max drop rate $g_{\text{max drop}}$	0.5 mm/mm
	Radius floor for check	0.55 mm
Radius floor	Absolute floor r_{floor}	0.25 mm
	Min segment length for floor	20.0 mm
Extreme narrowing	Max ratio $r_{\min}/r_{\text{ref}}$	0.25
	Min segment length	40.0 mm
Confidence gate	Min confidence $c_{\min}$	0.60

9 Domain Transfer to Clinical Data

A key requirement of this project is that the system, trained on publicly available ImageCAS data, must generalize to clinical CCTA scans from Ziv Medical Center. This section describes the challenges encountered and solutions developed for successful domain transfer.

9.1 Preprocessing Alignment

The single most important factor for domain transfer was strict alignment of preprocessing parameters between training and inference data. Two critical issues were discovered and resolved:

1. **Orientation mismatch**: ImageCAS data was preprocessed to RAS+ orientation, while Ziv data was initially processed to LAS orientation. Despite identical HU windowing, spacing, and normalization, this orientation mismatch caused complete model failure on clinical data the model produced empty predictions or noise. Reprocessing all Ziv data to RAS+ resolved the issue.

2. **HU window consistency**: Early experiments used varying HU windows (some training runs used $[-100, 1000]$, others used $[50, 450]$). Standardizing both datasets to $[-50, 700]$ ensured consistent tissue contrast representation.

3. **Field of view variability**: Unlike ImageCAS, where volumes are acquired with a cardiac-focused field of view (FOV) of approximately 150–250 mm, Ziv clinical scans lack a consistent acquisition protocol. Some cases use a narrow cardiac FOV comparable to the training data, while others are acquired with a wide FOV (300–350 mm) that captures the full chest. After resampling to 0.5 mm isotropic spacing, wide FOV volumes are substantially larger and exhibit different voxel level characteristics, as the same anatomical structures occupy proportionally fewer voxels. To address this, the preprocessing pipeline includes an automatic cardiac region-of-interest (ROI) cropping step. When the axial FOV exceeds a threshold of 280 mm, the algorithm localizes the heart by computing a mean intensity projection over the central 20% of z-slices (clipped to HU $[100, 700]$ to highlight contrast enhanced structures), and identifying the peak as the heart center. The volume is then cropped to a 220 mm target FOV centered on this point in the axial plane; the z-axis is preserved unchanged. This cropping is applied per axis only axes exceeding the threshold are cropped. Additionally, the DICOM-to-NIfTI series selection pipeline incorporates FOV into its scoring heuristic, preferentially selecting cardiac focused series when multiple series are available for the same patient.

9.2 Ziv Preprocessing Pipeline

A dedicated preprocessing pipeline was developed for Ziv data, handling the full conversion from DICOM format (the clinical standard at Ziv) to the Zarr format expected by the model. The pipeline performs:

1. **Series selection**: DICOM series are discovered, scored, and ranked using a heuristic that incorporates slice count, slice thickness, cardiac phase preference , FOV size, and series description keywords. Multi-phase series containing interleaved cardiac phases are penalized, and duplicate slices at identical z-positions are removed. When multiple table positions are merged under a single series number, z-gap detection splits them and retains the largest contiguous cluster.

2. **Conversion to NIfTI** via SimpleITK with z-position based slice ordering.

3. **Reorientation** to RAS+ using nibabel.

4. **Cardiac ROI cropping**: If the axial FOV exceeds 280 mm, the heart center is localized via intensity based peak detection and the volume is cropped to a 220 mm cardiac-focused FOV (as described in Section 9).

5. **Identical preprocessing** as ImageCAS: HU windowing, isotropic resampling, and z-score normalization with matching parameters.

6. **Storage** in Zarr format with Blosc compression and matching metadata structure.

9.3 Transfer Results

After preprocessing alignment, the segmentation model trained exclusively on ImageCAS produced anatomically plausible coronary vessel segmentations on Ziv clinical data without any site specific fine-tuning or domain adaptation. Three dimensional visualization confirmed recognizable coronary artery trees with appropriate branching patterns. This zero shot transfer validates the generalizability of the preprocessing model pipeline and suggests that consistent preprocessing including orientation standardization, HU window alignment, and FOV normalization is more impactful than domain adaptation techniques for this application.

10 Ground Truth Extraction from Clinical Reports

Ziv Medical Center provides structured radiology reports for each CCTA examination. These reports specify, for each coronary artery segment, the presence and severity of stenosis. To enable automated evaluation, we developed a PDF parser that extracts structured ground truth from these reports and outputs a standardized JSON file per patient. This JSON representation contains per artery records with fields for artery name, stenosis percentage, severity classification, and plaque information, providing a machine readable format that can be directly compared against the pipeline's own JSON output on a per artery and per severity basis.

10.1 Report Format

Each Ziv report contains findings organized by coronary artery (Left Main, LAD, LCx, RCA and their branches). For each artery, the report may specify the degree of stenosis (e.g., "mild," "moderate," "severe," or a percentage), and other clinical observations.

11 Evaluation

The system is intially evaluated at two levels: segmentation quality on ImageCAS, and stenosis detection accuracy against Ziv clinical ground truth.

11.1 Segmentation Evaluation

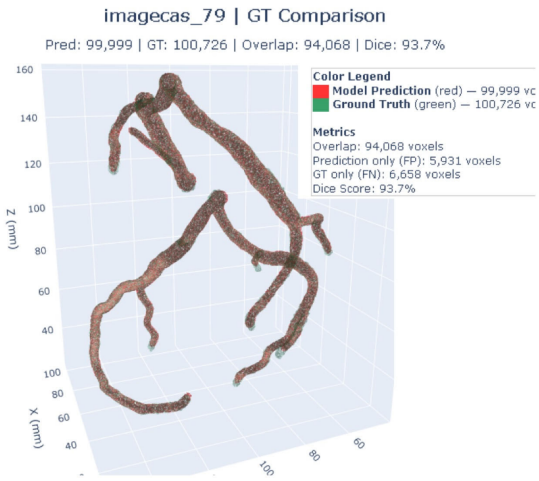
The primary segmentation metric is the Dice similarity coefficient:

$$\text{Dice} = \frac{2|P \cap G|}{|P| + |G|} \tag{17}$$

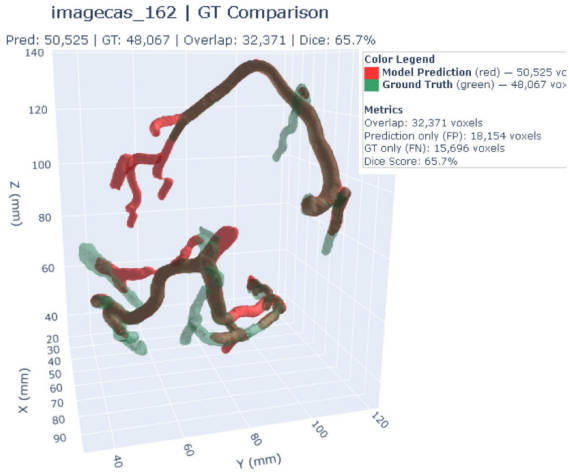
where P is the set of predicted vessel voxels and G is the ground truth. We also report precision $|P \cap G|/|P|$ and recall $|P \cap G|/|G|$ to characterize the segmentation behavior.

11.2 Qualitative Evaluation

Qualitative evaluation is performed through 3D visualization of the segmentation results, centerline extraction, and stenosis detections on both ImageCAS and Ziv data.

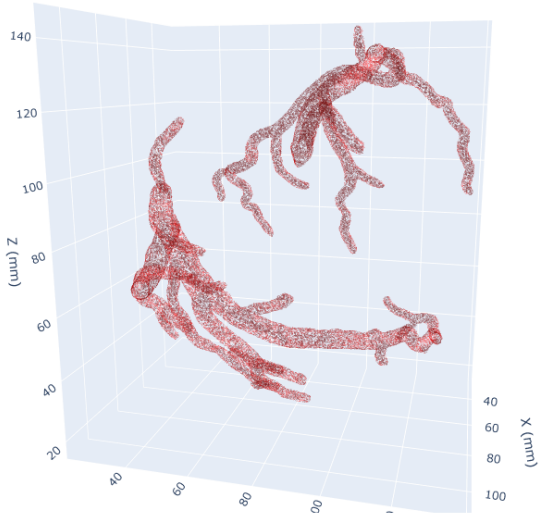


(a) ImageCAS case 79 (Dice 93.7%). Prediction (red) and ground truth (green) overlap closely, with minor false positives at distal branch tips.

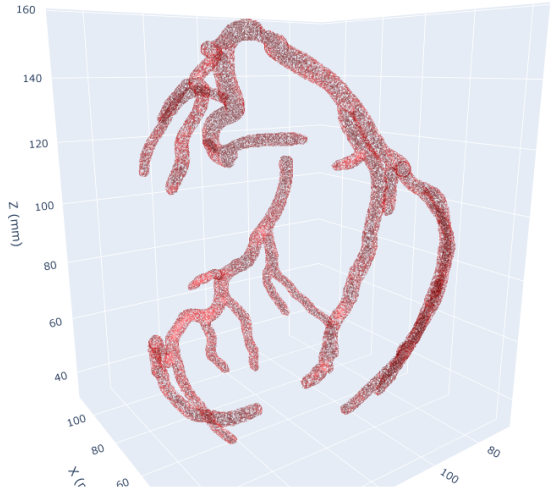


(b) ImageCAS case 162 (Dice 65.7%). The model over-segments in several regions (red clusters without green) and misses parts of the distal vessel tree (green without red).

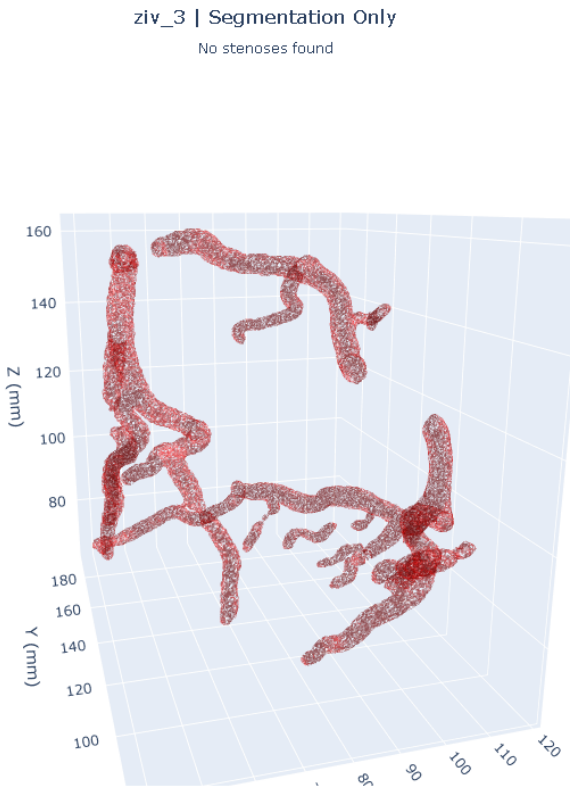
Figure 1: 3D segmentation results on ImageCAS test cases with ground truth overlay. Red: model prediction, green: ground truth, brown/overlap: agreement. (a) A well-segmented case representative of typical model performance. (b) A failure case showing both over-segmentation and missed distal branches.



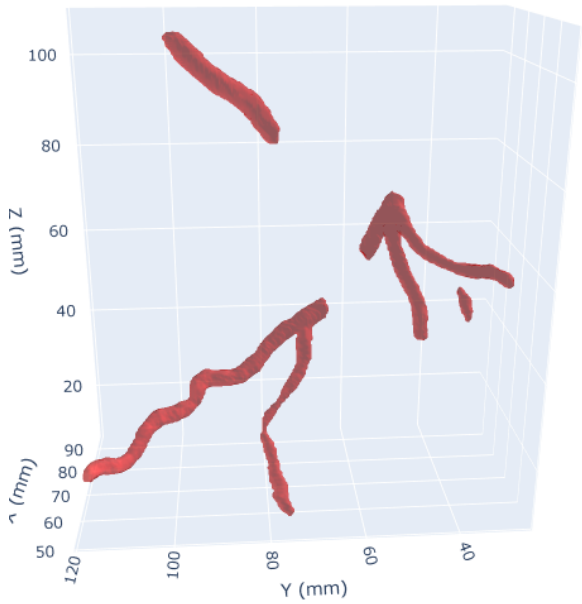
(a) Ziv case 30. Both left and right coronary systems are captured with anatomically plausible branching.



(b) Ziv case 16. Complete vessel tree with clear left/right separation and preserved distal branches.

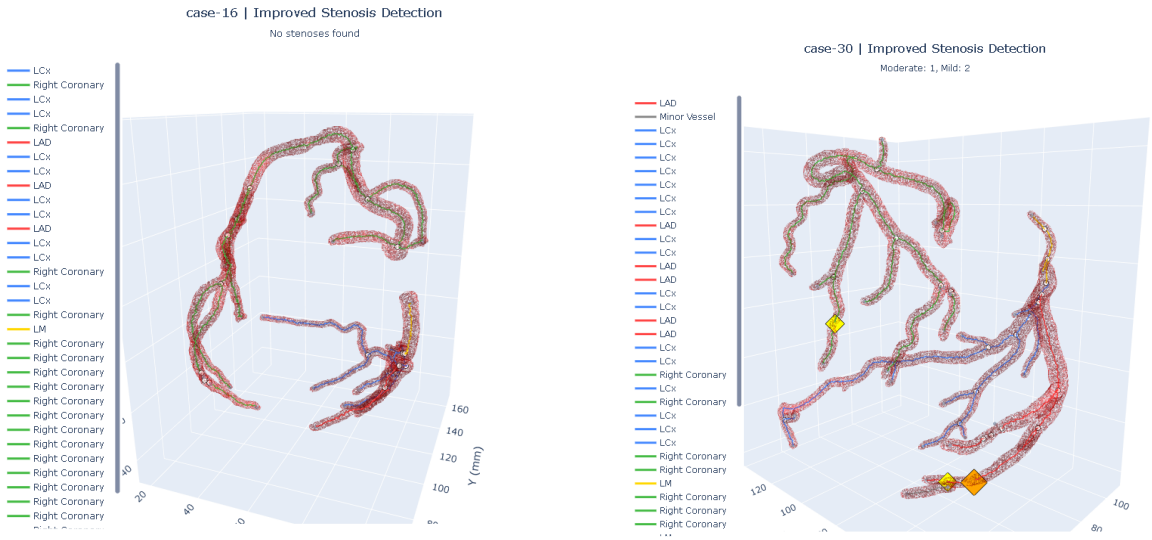


(c) Ziv case 3. Fragmented segmentation with disconnected vessel components and noisy small clusters.



(d) Ziv case 28. Severe under-segmentation producing only isolated vessel fragments with no connected tree structure.

Figure 2: 3D segmentation results on Ziv clinical cases (no ground truth masks available). Top row: cases where the model produces anatomically coherent coronary trees despite zero-shot domain transfer. Bottom row: failure cases where the model produces fragmented or incomplete segmentations, which propagate errors into downstream stenosis detection.



(a) Ziv case 16. The pipeline correctly reports no stenoses, matching the radiologist ground truth of normal findings across all vessels. Artery labels (LAD in red, LCx in blue, RCA in green, LM in yellow) are consistent with the expected anatomy.

(b) Ziv case 30. The pipeline detects one moderate and two mild stenoses. The radiologist ground truth reports one severe (LAD, 75%) and two mild findings. The severe LAD stenosis is correctly localized but under-graded as moderate, likely due to partial volume effects in the radius estimate at the narrowing site.

Figure 3: Stenosis detection results on two Ziv cases showing the complete pipeline output: 3D vessel segmentation with artery labels (color-coded by segment) and detected stenosis locations (diamond markers). (a) A true negative case. (b) A case illustrating the typical under-grading pattern where severe stenoses are detected at the correct location but classified one severity level below the radiologist assessment.

11.3 Quantitative Evaluation on Ziv Data

The stenosis detection pipeline is evaluated against radiologist ground truth on 28 Ziv Medical Center cases (cases 13 and 15 excluded due to missing data). Because the radiologist reports and the automated pipeline label vessels at different levels of granularity, two comparison strategies are used: a side-level comparison that aggregates all left and right coronary findings without requiring specific artery matching, and an artery-level comparison that evaluates the LM, LAD, LCx, and RCA independently.

For every comparison, each vessel (or side) is assigned a severity level in $\{\text{Normal}, \text{Mild}, \text{Moderate}, \text{Severe}\}$ based on its maximum stenosis percentage, following the thresholds in Table 4. True positives (TP) and false positives (FP) are computed against two binary decision rules:

- **Any stenosis**: severity $\geq$ Mild ($\geq 25\%$).
- **Significant stenosis**: severity $\geq$ Moderate ($\geq 50\%$).

Standard metrics—sensitivity, specificity, positive predictive value (PPV), accuracy, and F1 score—are reported for each rule. All results below are from the production configuration described in Sections 7 and 8, running on 28 Ziv cases preprocessed and segmented by the submission pipeline.

11.3.1 Side-Level Comparison

The side-level comparison aggregates all findings into Left Coronary (LM, LAD, LCx, and their branches) and Right Coronary (RCA and its branches) groups. For each case, the maximum

severity observed on each side from the pipeline output is compared against the maximum severity reported by the radiologist for that side. Exact severity agreement is achieved on 11/28 cases for both sides, meaning that for 39% of cases the pipeline and the radiologist assign the identical severity class (Normal, Mild, Moderate, or Severe) on a given side. Aggregate detection metrics are shown in Table 6.

Table 6: Side-level detection results on 28 Ziv cases. Patient-level significance is positive if either side has significant stenosis; side-level metrics treat each side as an independent case (56 total).

Metric	**TP**	**FP**	**TN**	**FN**	**Sens.**	**Spec.**	**F1**
Patient-level $\geq 50\%$	5	5	10	8	0.385	0.667	0.435
Side any stenosis $\geq 25\%$	12	17	14	13	0.480	0.452	0.444
Side significant $\geq 50\%$	4	6	33	13	0.235	0.846	0.296

The high specificity at the significant-stenosis threshold (84.6%) reflects the effectiveness of the artifact filters described in Section 8.5: the pipeline rarely produces false severe findings on normal patients. The lower sensitivity at the significant threshold (23.5%) is a consequence of the segmentation model systematically under-measuring the degree of narrowing at severely stenotic sites. Radiologist-reported severe stenoses (75–90%) are frequently measured by the pipeline at only 60–66%, which remains in the Moderate category under clinical severity thresholds.

11.3.2 Artery-Level Comparison

The artery-level comparison evaluates the pipeline's ability to localize stenosis to the correct specific coronary artery. Each case contributes four artery entries (LM, LAD, LCx, RCA), giving $28 \times 4 = 112$ total artery-level observations. A prediction labeled "Left Coronary" without a specific sub-artery designation, or labeled "Minor Vessel" because it was assigned to a small disconnected component, cannot be matched to a specific artery and is counted as unmapped. A total of 10 predictions are unmapped across the 28 cases, the majority of them from cases where the left coronary tree was too fragmented for the LM/LAD/LCx subdivision to complete.

Table 7: Per-artery detection results on 28 Ziv cases. "Any stenosis" uses the $\geq 25\%$ threshold; "Significant" uses the $\geq 50\%$ threshold. Exact match counts cases where the pipeline's severity class matches the radiologist's exactly (Normal/Mild/Moderate/Severe).

Artery	**Exact**	**TP**	**FP**	**TN**	**FN**	**Sens.**	**F1**
Any stenosis ($\geq 25\%$)							
LM	24/28	0	1	24	3	0.000	0.000
LAD	12/28	4	2	11	11	0.267	0.381
LCx	15/28	1	7	14	6	0.143	0.133
RCA	11/28	2	9	10	7	0.222	0.200
Overall any		7	19	59	27	0.206	0.233
Significant stenosis ($\geq 50\%$)							
LM		0	0	27	1	0.000	0.000
LAD		2	2	15	9	0.182	0.267
LCx		0	2	22	4	0.000	0.000
RCA		0	3	20	5	0.000	0.000
Overall sig.		2	7	84	19	0.095	0.133

Several observations follow from Table 7:

- **LAD achieves the best artery-level F1 (0.381)** on the any-stenosis threshold, reflecting the stability of the LAD/LCx split algorithm on the longest and most distinctive left-coronary vessel.
- **LM stenosis is almost never detected** because the ground truth contains only one significant LM finding (the remaining LM cases are Normal), and the pipeline's focus on the walk path through LM rarely produces stenosis findings on short trunk segments.
- **RCA has the highest false-positive rate** across all arteries. Empirical inspection of the failing cases showed that the RCA is systematically more prone to segmentation fragmentation than the left system, and several of the false positives are subtle narrowings in short RCA segments that the current filters do not catch.
- **10 unmapped predictions** (8.9% of all predictions) remain unassigned to a specific artery. Eliminating these would require either a more robust LAD/LCx subdivision on fragmented left systems, or a post-hoc mapping of "Minor Vessel" findings to the nearest significant component.

11.3.3 Severity Under-Estimation

The most consistent systematic error across both comparison levels is severity under-estimation. On the five Ziv cases with radiologist-reported severe stenosis ($\geq 70\%$), the pipeline's measured stenosis percentage is consistently 10–15 percentage points lower than the radiologist's reading, pushing many true-severe findings into the Moderate category. A concrete example is case 30: the radiologist reports LAD mid-segment stenosis of 75% (Severe), while the pipeline measures 66% (Moderate) on the same segment. The underlying cause is that the segmentation mask does not fully narrow at the stenotic site, likely because of partial volume effects in the presence of eccentric calcified plaque. Because the measured minimum radius r_{min} and the reference radius r_{ref} are both computed from the same mask, the ratio $r_{\text{min}}/r_{\text{ref}}$ systematically overestimates the true lumen ratio when the stenotic lumen is under-segmented.

This failure mode cannot be corrected by post-hoc parameter tuning alone: changing the reference percentile or smoothing sigma recovers at most 2–3 percentage points of stenosis measurement, insufficient to cross the Severe threshold for most affected cases. Reliable severe-stenosis detection on clinical data would require either (a) retraining the segmentation model with additional calcified-plaque cases, or (b) adding a complementary intensity-based narrowing measurement that does not depend on mask boundaries.

11.3.4 Impact of Artifact Filters

To quantify the value of the artifact filters introduced in Section 8.5, results were compared against an earlier pipeline configuration without any filters. The unfiltered baseline produced six false-severe findings on normal patients (21% false-positive rate at the patient level) driven by segmentation-gap artifacts. After adding the endpoint taper increase (6 mm), the gradient filter, the radius floor filter, the extreme narrowing filter, and the confidence gate, the patient-level false-severe rate dropped to the 5/28 reported in Table 6, and side-level specificity at the significant threshold improved from 70.0% to 84.6%. The filters did cost a small number of true positives (two previously-detected true severe findings fell below the 60% confidence gate), but the precision improvement more than compensates for the recall loss in terms of F1.

12 Demo Application

The full stenosis detection pipeline is packaged as an interactive desktop application to enable end-to-end processing on local hardware, without requiring cloud infrastructure or Jupyter note-

books.

12.1 Frontend

The application is implemented as a native PyQt desktop application (`app_qt.py`) using a horizontal split-pane layout. The window opens at 1600 × 900 pixels with a dark theme and the title "Stenosis Detection — Coronary Artery Analysis." A startup splash screen featuring an animated pulsing heart icon, a progress bar, and sequential status messages is displayed for a minimum of 2.5 seconds before fading out.

The left panel contains input controls: a folder browser for DICOM directories, NIfTI files, or preprocessed `.zarr` stores, an example case loader (pre-configured to `data/ziv/exampleCase.zarr`), and a path display field. Below these, a model selection combo box automatically discovers available checkpoints by scanning the `models/` directory for subdirectories containing `run_config.json` and a checkpoint file. Each entry displays the architecture type, loss function, and best validation Dice score. Two processing option checkboxes follow: a post-processing toggle (enabled by default) that controls segmentation mask cleanup, and an optional refinement toggle (disabled by default, adds approximately 10–30 seconds of processing time). Run and Cancel buttons control pipeline execution, with a Clear button to reset all results. An eight-step pipeline stepper widget displays per-stage progress using color-coded icons: gray circles (waiting), blue gears (running), green checkmarks (done), and red crosses (failed). A log viewer displays detailed pipeline messages in real time.

The right panel hosts a five-tab results viewer and a log panel. All pipeline stages are orchestrated by a shared runner (`runner.py`) that executes in a background `QThread`, communicating progress to the UI via Qt signals. Two additional worker threads handle asynchronous tasks: a `ZarrLoaderWorker` for loading preprocessed data without blocking the UI, and a `RegionMaskWorker` for computing vessel region masks between user-selected 3D points.

12.2 Results Tabs

The results panel provides five tabs for inspecting pipeline output:

1. **3D Visualization**: An interactive Plotly 3D vessel rendering embedded in a `QWebEngineView`, with vessels colored by artery system and stenosis findings shown as severity-colored spheres (red for severe, orange for moderate, yellow for mild, green for normal). A `PlotlyBridge` module bridges JavaScript click events to Qt signals via `QWebChannel`, converting Plotly millimeter coordinates to voxel indices. Clicking a stenosis marker automatically navigates to that location in the 2D slice viewer. A region selection mode allows the user to click two points on a vessel in 3D; the application then computes a centerline-based region mask between those points and highlights it in cyan on the 2D viewer. Three control buttons are provided: Select Region, Clear Region, and Reset Camera.

2. **2D Slice Viewer**: A `pyqtgraph`-based multi-planar reconstruction (MPR) viewer supporting single-axis mode (Axial, Coronal, or Sagittal with a slice slider) and an All Views mode displaying synchronized three-panel axial/coronal/sagittal views with draggable crosshairs. The viewer composites five rendering layers: grayscale CT image, semi-transparent vessel mask overlay, cyan region highlight (from 3D region selection), severity-colored finding markers (× symbols scaled by severity), and crosshair lines. Five window presets are available for contrast adjustment (Lung, Mediastinum, Bone, Brain, Soft Tissue). A finding dropdown enables jump-to-finding navigation, and independent zoom and pan are supported per panel with automatic view fitting on view switches.

3. **Clinical Report**: A formatted multi-section plain-text report generated by `runner.py`. The report includes a header with case identifier and processing time, a summary section

with segment count, total findings, and severity distribution, a detailed findings section listing each detection sorted by severity (with artery name, segment ID, stenosis percentage, minimum and reference radii, voxel location, and confidence score), an impression section with clinical interpretation based on maximum detected severity, and a disclaimer noting the output is automated and should be validated by qualified medical professionals.

4. **Raw Data (JSON)**: The complete pipeline output in structured JSON format, containing a summary object (segments analyzed, total findings, severity distribution), a findings array (with per-finding fields for segment ID, artery name, region, severity, stenosis percentage, radii, voxel location, and confidence), and visualization data (vessel centerlines, radii, segment identifiers). This format supports programmatic inspection and downstream analysis.

5. **GT Comparison**: A side-by-side comparison of automated findings against radiologist ground truth, displayed when a matching clinical report is available. The application discovers ground truth reports either as sibling JSON files alongside `.zarr` inputs or via a `ziv_manifest.json` lookup. The comparison module (`compare_gt.py`) performs per-artery and per-side matching with canonical artery name normalization, computes true positives, false positives, and false negatives, and displays segment-by-segment severity agreement in a tabular format.

12.3 Pipeline Stages

The application runs the following eight stages in sequence, with per-stage progress updates displayed in the stepper widget:

1. **DICOM to NIfTI**: Discovers DICOM series from an input folder, scores and selects the appropriate CCTA series using a heuristic incorporating slice count, slice thickness, cardiac phase preference, and FOV size, and converts to a single NIfTI volume via SimpleITK.

2. **Preprocessing**: Applies HU windowing $[-50, 700]$, isotropic resampling to 0.5 mm, reorientation to RAS+, and z-score normalization. Output is written to Zarr format.

3. **Segmentation**: Loads the selected model checkpoint, runs sliding window inference (patch size 160^3, 50% overlap, Gaussian weighting), and writes float16 probability maps alongside the binary mask.

4. **Post-processing**: Runs the distance-based fragment removal pipeline (Section 5) on the binary mask. This stage is skippable via the post-processing checkbox.

5. **Centerline extraction**: Extracts the vessel skeleton, builds the segment graph, and estimates smoothed radius profiles.

6. **Artery labeling**: Classifies segments as Left Coronary, Right Coronary, or minor vessel using connected-component analysis (Section 7).

7. **Stenosis detection**: Computes per-segment stenosis scores and severity classifications (Section 8).

8. **Visualization**: Generates an interactive 3D HTML visualization using Plotly, with vessels colored by artery system and stenosis findings shown as labeled markers.

For preprocessed `.zarr` inputs, stages 1 and 2 are automatically skipped based on input type detection. For NIfTI inputs, only stage 1 is skipped. If any stage fails, the pipeline halts immediately with an error message displayed in a dialog and the failing stage marked red in the stepper.

12.4 Hardware and Performance

The application targets the following local hardware configuration: mboxNVIDIA GeForce GTX 1650 (4 GB VRAM), AMD Ryzen 7 5800H (8 cores), 32 GB RAM. Sliding window inference with 160^3 patches fits within the 4 GB VRAM limit; if GPU memory is insufficient, the application automatically falls back to CPU inference.

Table 8 summarizes the per-stage processing times. End-to-end processing time for a representative Ziv test case (case-30) is approximately 84.5 seconds on the above hardware, encompassing all eight pipeline stages. Typical total processing time ranges from 80 to 170 seconds depending on input format (DICOM requires conversion overhead), volume size, and whether optional post-processing and refinement are enabled.

Table 8: Per-stage processing times on local hardware (GTX 1650, Ryzen 7 5800H).

Stage	Operation	Duration
1	DICOM to NIfTI	10–30 s
2	Preprocessing	20–40 s
3	Segmentation	30–60 s
4	Post-processing	1–2 s
5	Centerline extraction	5–10 s
6	Artery labeling	3–5 s
7	Stenosis detection	5–10 s
8	Visualization	5–10 s
	Total (typical)	**80–170 s**

12.5 Ground Truth Comparison

A dedicated comparison module (`compare_gt.py`) loads the structured JSON findings produced by the pipeline alongside parsed clinical report ground truth, and computes per-artery and per-side severity metrics. The module normalizes artery names to canonical forms (LM, LAD, LCx, RCA), maps arteries to coronary sides (Left, Right), and evaluates at both per-artery and per-side granularity. A positive threshold of $\geq 25\%$ stenosis defines clinically significant findings. The application exposes a comparison tab where the user can immediately see the segment-by-segment severity agreement between the automated findings and the clinical ground truth.

13 Conclusion

This project presented an automated system for detecting and quantifying coronary artery stenosis from CCTA scans, built around a two-stage design: supervised deep learning for vessel segmentation, followed by unsupervised computational geometry for stenosis measurement. The deliberate separation of these stages makes every stenosis finding traceable to specific radius measurements along the vessel centerline, avoiding the opacity of end-to-end black-box approaches.

The segmentation model, a 3D U-Net trained on the ImageCAS dataset, achieves 82.4% Dice on held-out ImageCAS cases and transfers to Ziv Medical Center clinical data without any site-specific fine-tuning. The key enabler for this domain transfer was strict preprocessing alignment particularly orientation standardization to RAS+ and consistent HU windowing rather than architectural complexity or domain adaptation techniques. An orientation mismatch between training and clinical data caused complete model failure before this was identified and corrected, underscoring that data configuration decisions can have larger impact than model design choices.

The geometric analysis pipeline centerline extraction, artery labeling via junction-graph topology, and segment-based stenosis quantification produces clinically formatted findings that

can be directly compared against radiologist reports. On 28 Ziv cases, the system achieves 84.6% specificity at the significant-stenosis threshold after artifact filtering, meaning it rarely produces false severe findings on normal patients. Sensitivity remains lower (23.5% at the significant threshold), driven primarily by systematic severity under-estimation: the segmentation mask does not fully narrow at stenotic sites in the presence of calcified plaque, causing the measured stenosis to fall 10–15 percentage points below the radiologist reading. This is a segmentation-level limitation that cannot be resolved by downstream parameter tuning alone.

Several lessons from this project extend beyond the specific application. First, analyzing the characteristics of the processed training data rather than the raw scans proved essential for making correct preprocessing decisions. Second, mask resampling with linear interpolation and a 0.3 threshold prevented 82% vessel voxel loss that nearest-neighbor interpolation caused on thin tubular structures—a finding relevant to any segmentation task involving elongated geometry. Third, artifact filters grounded in domain knowledge (radius gradients, anatomical radius floors) were more effective at reducing false positives than generic post-processing, because they target the specific failure modes of the upstream segmentation model.

The main limitations of the current system are the severity under-estimation discussed above, the reliance on segmentation quality for all downstream measurements, and the 8.9% rate of unmapped artery-level predictions caused by vessel fragmentation. Addressing these would require either retraining the segmentation model on data that better represents calcified-plaque cases, or introducing a complementary intensity-based narrowing measurement that does not depend on mask boundaries. The artery labeling could be further improved by extending the junction-graph reconnection to handle more severe fragmentation patterns, or by incorporating learned anatomical priors.

The complete pipeline is deployed as an interactive desktop application that processes a CCTA scan end-to-end in 80–170 seconds on consumer hardware (GTX 1650, 4 GB VRAM), producing 3D visualizations, 2D slice overlays, and structured clinical reports. All source code, model checkpoints, and evaluation scripts are provided in the accompanying repository.

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